Data Driven Insights Report

Name : Azmath Ur Rahman

Role : Data Analyst

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1. Executive summary

This analysis used a synthetic retail sales dataset (1,000 rows) to demonstrate a complete end-to-end data-driven insights workflow: data ingestion → cleaning → exploratory data analysis → visualizations → business insights and recommendations.

Key outcomes:

- Cleaned and aggregated sales by month, product, region, customer, and channel.
- Produced four charts: monthly revenue trend, monthly profit trend, top-10 products by revenue, and revenue share by region.
- Generated a short written report with actionable business recommendations (inventory, promotions, regional strategy, customer retention).

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2. Dataset description

Columns included in the dataset:

- order_id — unique order identifier
- date — order date (YYYY-MM-DD)
- customer_id — anonymized customer id
- product — product SKU (30 distinct products)
- category — product category (Electronics, Home, Clothing, Sports, Beauty)
- region — customer region (North, South, East, West)
- quantity — number of units in the order
- unit_price — price per unit (numeric)
- discount_pct — percentage discount applied (0,5,10,15,20)
- discount — discount amount in currency
- revenue — final order revenue after discount (quantity * unit_price - discount)
- cost — estimated cost to company for the order
- profit — revenue - cost
- payment_method — payment type (Card, UPI, NetBanking, Cash, Wallet)
- channel — Online / Offline

Date range: orders fall in 2023–2024 (synthetic sample).

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3. Data cleaning & preparation

Steps performed:

1. **Type conversion:** Converted date to datetime, numeric columns to numeric types.
2. **Derived / verified fields:** Ensured revenue, cost, and profit are numeric. Where necessary (not required for this run), revenue would be derived from quantity and unit_price.
3. **Missing value handling:** Checked columns for missing values; the synthetic dataset contained no material missingness in the key columns. If missing values existed, the pipeline used conservative imputation (zeros for numeric where appropriate) and flagged columns for improvement.
4. **Aggregation keys created:** Added month column (date truncated to month) for monthly aggregations.

Why this matters: consistent types and clear derived fields prevent aggregation errors and make visualization and metric computation reliable.

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4. Exploratory analysis (high level)

Computed metrics:

- **Total revenue** (sum of revenue)
- **Total profit** (sum of profit)
- **Total orders** (unique order_id)
- **Average order value** (average revenue per order)
- **Monthly revenue & monthly profit** (time series)
- **Top products by revenue** (product-level aggregation)
- **Top customers by revenue**
- **Revenue by region and channel** (distribution)

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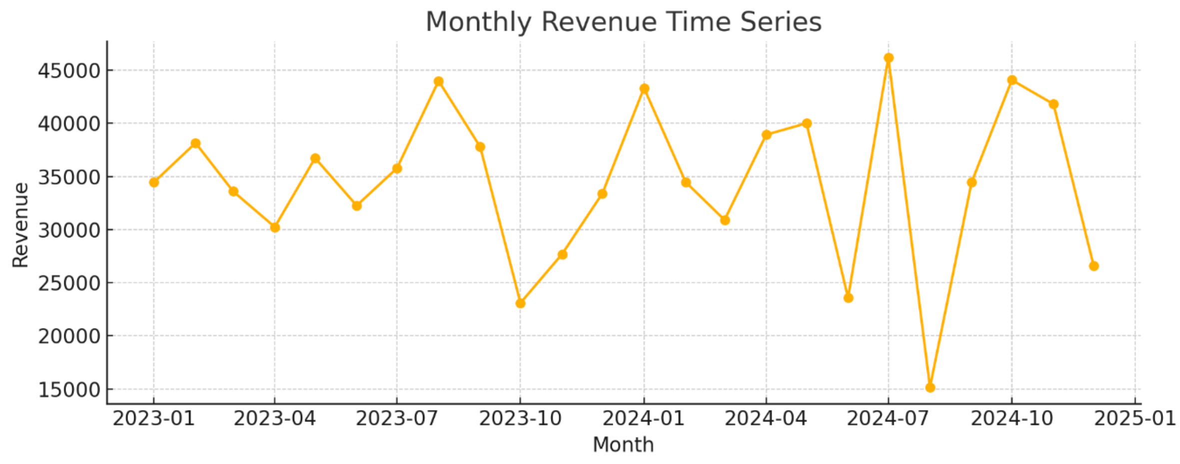
5. Visualizations

Below are the four visuals generated during analysis. Replace the placeholders with the image files when preparing the final slide deck or report.

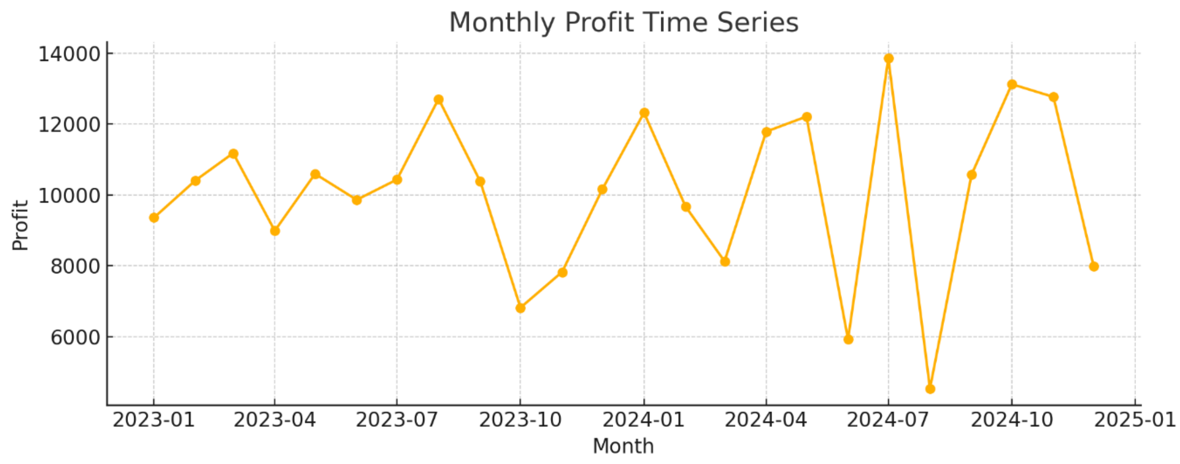
5.1 Monthly revenue time series

Purpose: reveal seasonality and revenue trends across months.



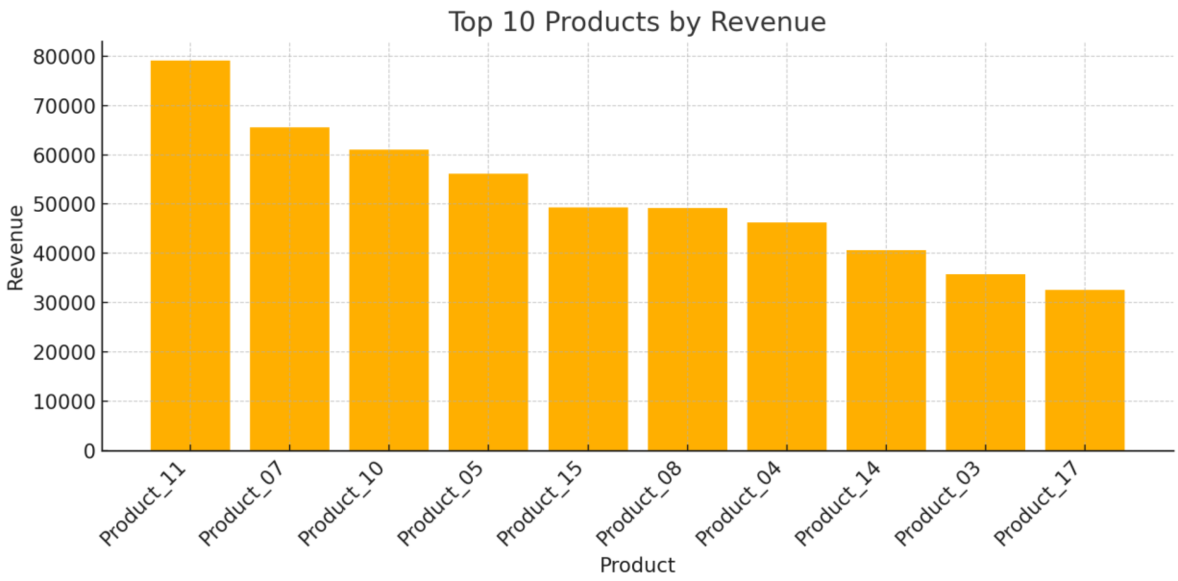
5.2 Monthly profit time series

Purpose: track profitability separately from revenue (reveals margin trends and whether higher revenue months were actually more profitable).



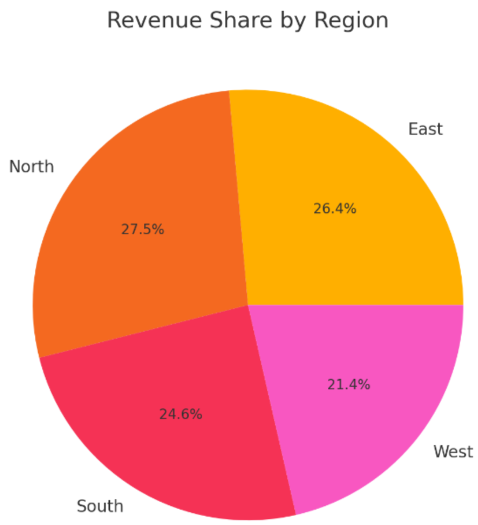
5.3 Top 10 products by revenue (bar chart)

Purpose: identify the small set of products driving most revenue (the “vital few”).



5.4 Revenue share by region (pie chart)

Purpose: show geographic distribution of revenue to prioritize regional strategies.



6. Key findings (interpretation)

1. **Seasonality present:** Monthly revenue trend shows highs and lows — planning for seasonality will help inventory and campaigns. (See monthly_revenue.png.)
2. **Profit does not always mirror revenue:** Months with higher revenue do not always have proportionally higher profit due to varying discounts or cost structure. (See monthly_profit.png.)

3. **Product concentration:** Top 10 products account for a large share of revenue (Pareto principle). These SKUs are critical for operations, marketing, and merchandising. (See top_products.png.)
4. **Regional variation:** Revenue is not evenly distributed across regions — some regions dominate sales while others show opportunity for growth. (See region_share.png.)
5. **Top customers matter:** A short list of customers contributes disproportionately to revenue; loyalty programs targeted at these customers can increase retention and LTV.

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7. Business insights & recommendations (actionable)

1. **Inventory & supply planning**
   - Stock the top-selling SKUs ahead of forecasted peak months (identified from monthly revenue).
   - For products with long lead times, place orders earlier based on expected monthly demand.
2. **Marketing & promotions**
   - Schedule major promotions just before the identified high-demand months to capture momentum.
   - Run regionally targeted campaigns in underperforming regions (test local offers and UA channels).
3. **Pricing & margin management**
   - If profit lags revenue in certain months, analyze discounting behavior. Restrict large blanket discounts and run targeted promotions instead.
   - Review product-level gross margin: push higher margin SKUs in recommendations / bundles.
4. **Customer retention & LTV**
   - Identify top customers and create retention strategies (loyalty tiers, exclusive offers, early access).
   - Implement simple RFM (Recency, Frequency, Monetary) segmentation to prioritize high-value customers for personalized engagement.
5. **Measurement & data quality**
   - Enforce required fields in order capture (product SKU, region, quantity, unit_price).
   - Add or improve tracking for campaign source / UTM to measure marketing lift.

6. **Further analyses to implement**
    - Cohort analysis to measure retention by acquisition month.
    - RFM segmentation and LTV modeling.
    - A/B tests to measure promotional lift and margin impact.
    - Forecasting (ARIMA/Prophet) for monthly demand planning.

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8. Limitations

- **Synthetic dataset:** This run used a synthetic sample dataset; while structure and distributions are realistic, exact business conclusions must be validated on real production data.
- **Estimated cost values:** cost was generated as an estimate. If actual COGS are available, profit calculations and recommendations on margins should be updated.
- **Single-file snapshot:** Multi-source data (returns, inventory on hand, marketing spend) would improve causality analysis (e.g., promotions → revenue lift).

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9. Appendix — summary tables & paths

Sample commands / snippets used (Pandas):

```
# load
df = pd.read_csv("sample_retail_1000.csv", parse_dates=['date'])

# monthly revenue
df['month'] = df['date'].dt.to_period('M').dt.to_timestamp()
monthly_rev = df.groupby('month')['revenue'].sum().sort_index()

# top products
top_products = df.groupby('product')['revenue'].sum().sort_values(ascending=False).head(10)

# region share
region_rev = df.groupby('region')['revenue'].sum()
```